

HealthConnect Clinic Experience Lab

Initial HealthConnect Analytics Report

Exploratory Analysis, KPI Development & Business Insights

Executive Summary

The Health Connect appointment dataset contains 5,000 appointment records. Among appointments that reached an attendance outcome (Attended or No-Show), the overall no-show rate is 48.46%. The strongest initial patterns are associated with booking lead time and previous no-show history. Appointments booked 46-60 days in advance recorded a 53.82% no-show rate, compared with 27.81% for appointments booked within 0-7 days.

The dataset was prepared without overwriting or removing original records, key quality checks were completed, five business KPIs were calculated, exploratory analysis was conducted across appointment, patient, historical, reminder and operational characteristics, and an initial analytical dashboard was created.

The analysis also found that appointments receiving reminders had a lower observed no-show rate (47.36%) than appointments without reminders (51.39%), and no-show rates generally increased with distance from the clinic. These findings are descriptive associations and should not be interpreted as proof of causation.

The main business recommendations are to prioritise long-lead appointments and patients with prior no-show history for proactive confirmation, review reminder coverage, investigate reminder-channel performance further, and consider administrative support for patients travelling longer distances.

Business Problem and Analytical Objective

HealthConnect Clinic seeks to reduce missed appointments, better understand attendance patterns, improve utilisation of appointment slots and provide more effective administrative support. From the Data Analytics perspective, the Week 5 objective is to quantify appointment outcomes and identify meaningful patterns associated with no-shows that can support operational decision-making.

Business Questions

1. What proportion of eligible appointments result in no-shows?
2. Does booking lead time relate to appointment attendance?
3. Does previous no-show history identify appointments with a higher observed no-show rate?
4. Do reminder status and reminder channel show different attendance patterns?
5. Does distance to the clinic appear to be associated with no-show rates?
6. Do appointment type, day and time show meaningful differences in no-show rates?
7. Do patient characteristics such as age group and gender show meaningful differences?
8. How should cancellations be separated from no-shows when calculating attendance KPIs?

Project Resources and Dataset Overview

The analysis used the HealthConnect Appointment Dataset and HealthConnect Data Dictionary. The Data Dictionary was used to confirm variable meaning, expected data types and important notes such as patient

IDs appearing across multiple appointments, booking lead days being derivable, reminder channel being None when no reminder is sent, and limited missing values in distance and waiting time.

Item	Value
Records	5,000
Variables	18 original variables
Unique appointment IDs	5,000
Unique patients	1,696
Appointment date range	2025-01-01 to 2026-06-30
Booking date range	2024-11-07 to 2026-06-27
Appointment outcomes	Attended, No-Show, Cancelled

Variable Group

Variable Group	Variables
Identifiers	appointment_id, patient_id
Demographics	gender, age, age_group
Appointment characteristics	appointment_type, appointment_date, appointment_day, appointment_time
Booking	booking_date, booking_lead_days
History	previous_appointments, previous_no_shows
Reminders	reminder_sent, reminder_channel
Operational/access	distance_to_clinic_km, waiting_time_minutes
Outcome	appointment_outcome

Data Preparation and Data Quality

The original dataset was preserved. A separate processed dataset was created for analysis. Dates were converted to date types, numerical fields were validated, and derived analytical fields were added for segmentation. No original field was overwritten.

Data Quality Assessment

Check	Result	Status	Action / Interpretation
Record count	5,000	Pass	All records retained
Column count	18	Pass	Matches project dataset structure
Unique appointment IDs	5,000	Pass	No duplicate appointment IDs
Exact duplicate rows	0	Pass	No exact duplicates detected
Missing distance_to_clinic_km	90 (1.8%)	Review	Retained as missing; grouped as Unknown for segmented analysis
Missing waiting_time_minutes	60 (1.2%)	Review	Retained as missing; grouped as Unknown for segmented analysis
Booking date after appointment date	0	Pass	No chronological errors
Booking lead-day mismatches	0	Pass	Lead days agree with date difference
Appointment-day mismatches	0	Pass	Day labels agree with dates
Age-group mismatches	0	Pass	Age group agrees with age
Previous no-shows > previous appointments	0	Pass	Historical counts are logically consistent
Reminder channel logic errors	0	Pass	None is consistently used where no reminder was sent

Data Preparation

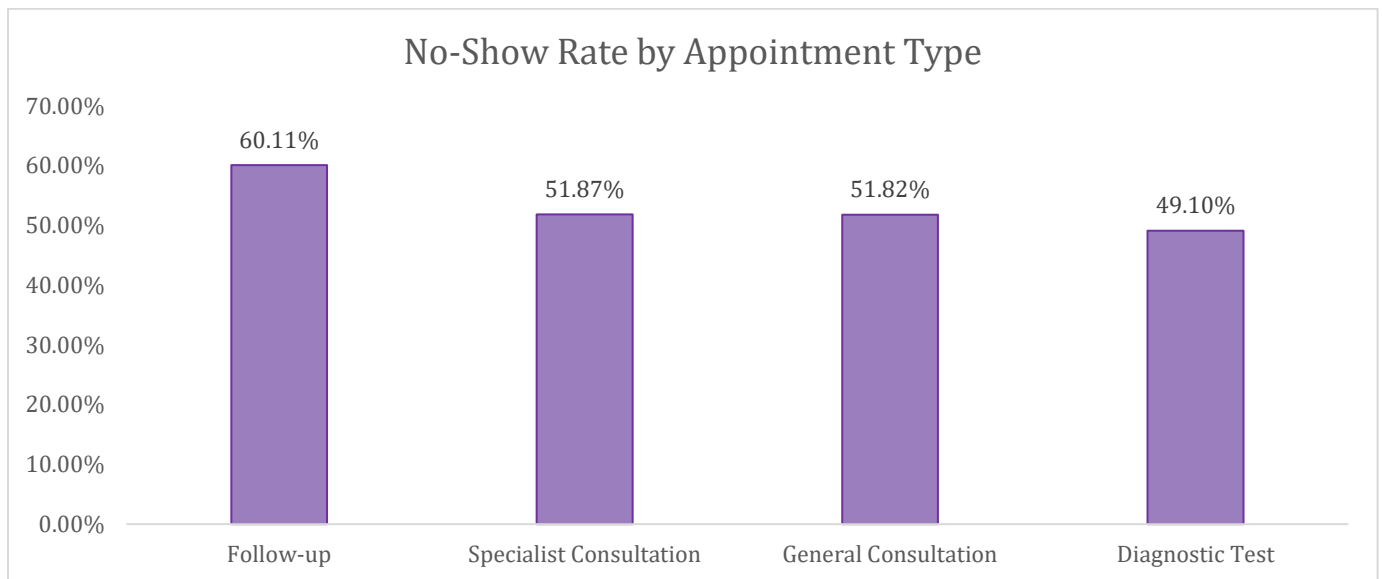
- No rows were deleted because no duplicate or structurally invalid records were found.
- Missing distance and waiting-time values were retained as null rather than replaced with zero.

- The reminder_channel value None was retained as a valid category rather than treated as a missing value.
- Derived fields were created for appointment month, booking lead-time band, previous no-show group, distance band and waiting-time band.
- Attended and No-Show appointments were flagged as eligible for attendance-rate analysis; Cancelled appointments were treated separately.

Exploratory Data Analysis

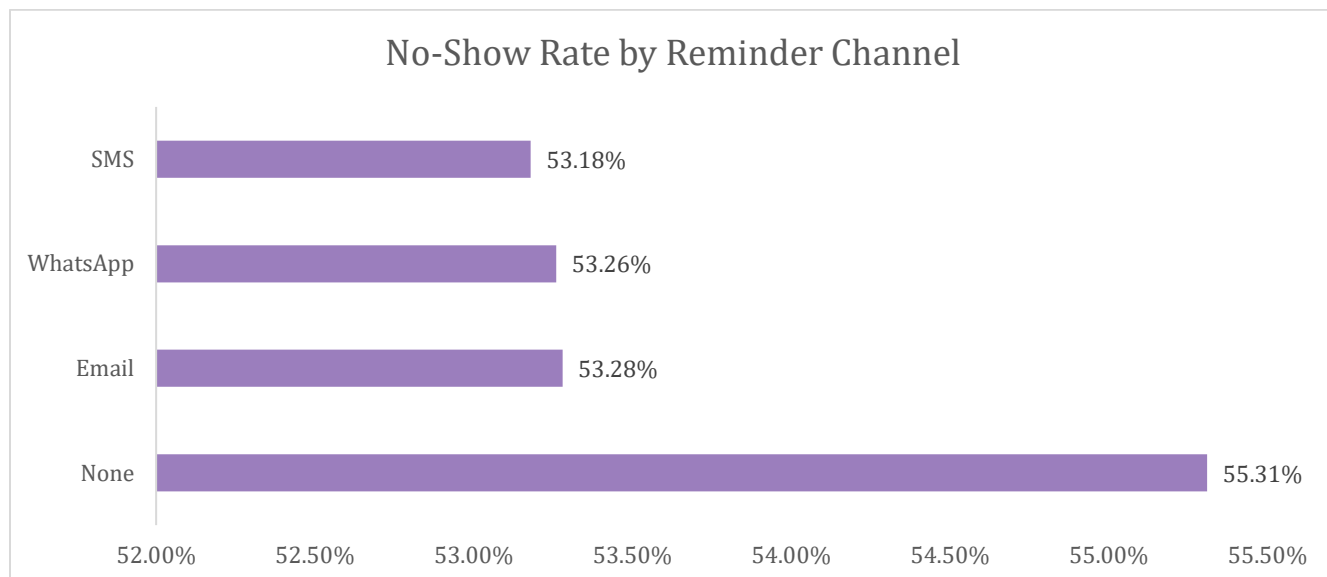
Appointment Outcomes

The dataset contains 2,423 No-Show appointments, 2,314 Attended appointments and 263 Cancelled appointments. No-Shows are therefore the most frequent final outcome in the dataset.

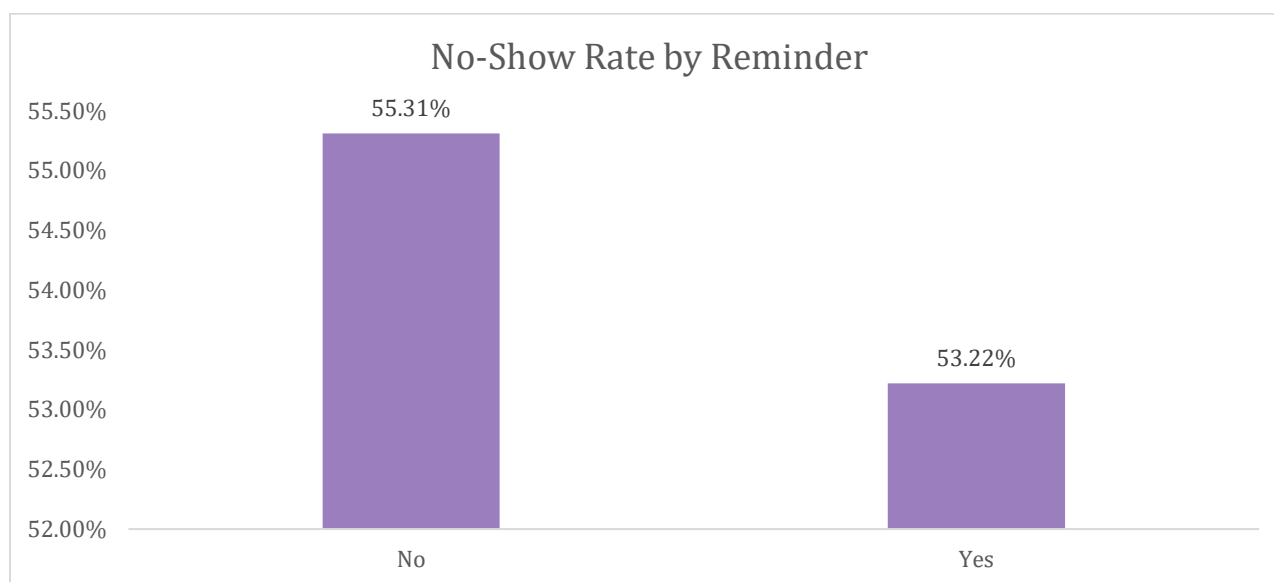


Appointment Characteristics

appointment_type	Total	No-Show Rate
General Consultation	2086	51.82%
Follow-up	1421	60.11%
Specialist Consultation	900	51.82%
Diagnostic Test	593	49.10%



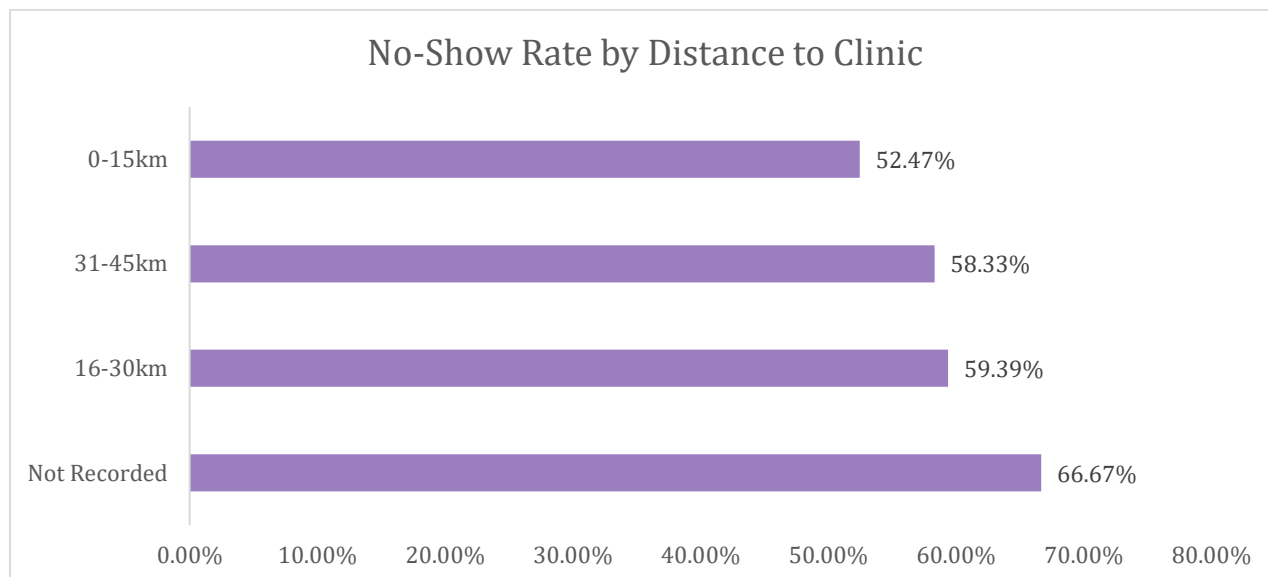
SMS has the lowest observed no-show rate among the available channel categories (53.18%). Because the dataset is observational and reminder channels were not randomly assigned, these results should be interpreted as associations rather than causal effects.



Appointments with a reminder have an observed no-show rate of 53.22%, compared with 55.31% for appointments without a reminder.

6.5 Distance to Clinic

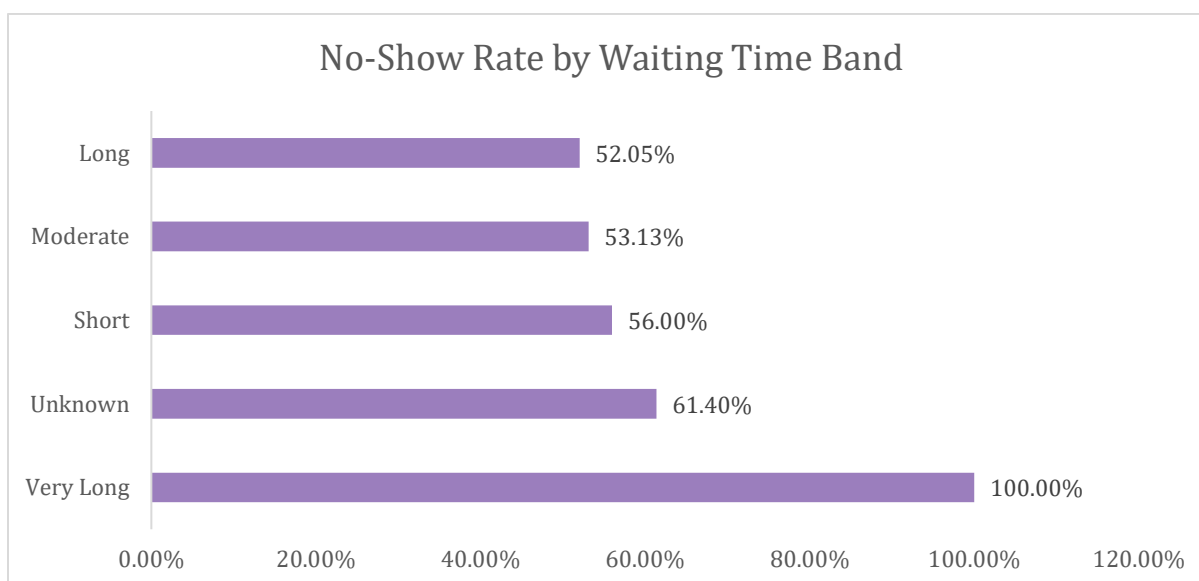
distance_band	Total	No-Show Rate
0-15 km	3980	52.47%
16-30 km	858	59.39%
31-45km	72	58.33%
Not Recorded	90	66.67%



No-show rates generally increase with distance: 52.47% at 0-15 km, 59.39% at 16-30 km, 58.33% at 31-45 km. The 31-45 km group contains only 72 appointments, so the 58.33% result should be interpreted cautiously.

6.6 Waiting Time

waiting time band	Total	No-Show Rate
0-15 min (Short)	1073	56.00%
16-30 min (Moderate)	2472	53.13%
31-45 min (Long)	1260	52.05%
46+ min (Very Long)	3	100.00%
Unknown	192	61.40%

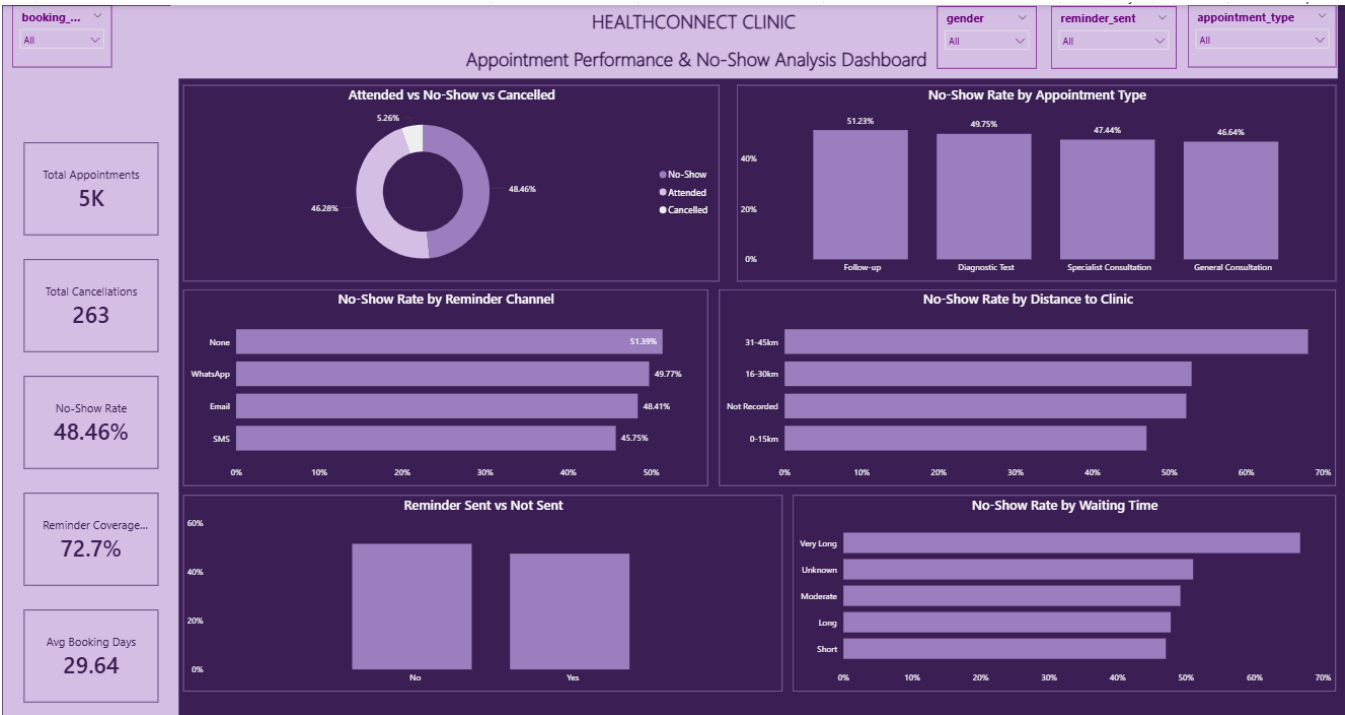


KPI

Five KPIs were selected. Each KPI is linked to a business question and interpreted below. For No-Show Rate and Attendance Rate, Cancelled appointments are excluded from the denominator because cancellation is analytically distinct from failing to attend.

Analytical Dashboard

A PowerBI analytical dashboard was created as a supporting file. The dashboard contains the five selected KPIs plus high-priority visuals for booking lead time, previous no-show history, reminders, distance and the monthly trend.



Key findings

1. Booking lead time shows the strongest initial pattern: no-show rate rises from 27.81% for appointments booked 0-7 days ahead (short notice) to 53.82% for appointments booked 46-60 days ahead (1-1.5 months).
2. Previous no-show history is strongly associated with current no-show behaviour: 46.30% with no previous no-shows versus 70.33% with three or more.
3. Appointments with reminders have a lower observed no-show rate (47.36%) than appointments without reminders (51.39%). This is an association, not proof of causation.
4. No-show rates generally increase with distance from the clinic, reaching 52.91% for 16-30 km and 68.06% for the 30+ km group. The 30+ km group is small (72 appointments), so the result should be interpreted cautiously.
5. Follow-up appointments have the highest observed no-show rate among appointment types at 51.23%.

Business recommendations

- Prioritise long-lead appointments for proactive confirmation or enhanced reminder workflows.
- Use previous no-show history as a segmentation factor for targeted administrative follow-up.
- Review why 51.39% of appointments did not receive a reminder and assess whether coverage can be improved.
- Investigate reminder-channel performance further before attributing lower no-show rates to a specific channel.
- Consider additional administrative support for longer-distance patients, subject to further analysis and clinic policy.
- Review follow-up appointment workflows because this appointment type has the highest observed no-show rate.

Assumptions, Limitations, Risks and Dependencies

Type	Issue	Impact / Response
Data limitation	Dataset is fictional/synthetic	Findings are project insights, not real clinical evidence.
Missing data	90 distance values and 60 waiting-time values are missing	Reduces available observations for those segmented analyses.
Reminder information	Reminder sent does not confirm delivered/read	Limits claims about reminder effectiveness.
Observational design	Relationships are not causal	Use association language and avoid causal recommendations.
Repeated patients	1,696 unique patients across 5,000 appointments	Appointment records are not fully independent observations.
Small segment	31-45 km group contains 72 appointments	Interpret the 68.02 % no-show rate cautiously.
KPI definition	Cancellation treatment changes denominators	Maintain the documented rule consistently across report and dashboard.
Unobserved factors	Transport, illness, socioeconomic circumstances and reasons for non-attendance are unavailable	The analysis cannot explain all causes of no-shows.
Dependency	Week 5 cross-track interaction must be completed	Document actual information exchange and impact before final submission.

Conclusion

The dataset is sufficiently structured for exploratory analysis, with limited missing operational values and no duplicate or major consistency problems identified. Five KPIs were calculated and an initial analytical dashboard was produced.

The most decision-relevant initial findings are the strong increase in no-show rate with longer booking lead times and the higher no-show rates among patients with previous no-show history. Reminder use and distance also show meaningful associations. These results support targeted operational follow-up while remaining subject to the limitations of a synthetic observational dataset.

Before final submission, the intern must complete and document the required cross-track collaboration and replace the author/date placeholders.